WiCaP: Reliable WiFi-Based Human Activity Recognition Under Environment Shift via Weighted Conformal Prediction

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***Abstract*—WiFi-based human activity recognition (HAR) has emerged as a promising sensing paradigm for applications including smart homes, occupancy monitoring, healthcare, and assisted living. Despite substantial progress in deep learning-based WiFi sensing, existing systems are typically evaluated using classification accuracy under controlled experimental conditions. In practical deployments, however, environmental changes such as room layout variation, user diversity, hardware differences, and multipath propagation shifts often induce significant distribution shift between training and deployment conditions. Under such shifts, modern neural networks may produce highly confident but incorrect predictions, creating reliability and safety concerns for real-world sensing applications.**

In this work, we investigate uncertainty-aware WiFi sensing through the lens of conformal prediction. We propose WiCaP, a reliability-oriented framework that combines domain-invariant representation learning, conformal calibration, and selective prediction for WiFi-based human activity recognition. WiCaP consists of three components: (1) a Domain-Contrastive Encoder (DCE) that learns activity-discriminative and environment-invariant representations using supervised contrastive learning and adversarial environment decorrelation; (2) a Conformal Calibration Layer (CCL) that converts classifier outputs into prediction sets with statistically valid coverage properties; and (3) a Selective Prediction Module (SPM) that leverages conformal uncertainty estimates to abstain on uncertain inputs. In addition, we introduce the Silent Failure Rate (SFR), a reliability metric that quantifies the frequency of high-confidence incorrect predictions under distribution shift.

Unlike conventional WiFi sensing approaches that focus solely on classification accuracy, WiCaP emphasizes reliability, calibration quality, coverage validity, and uncertainty-aware decision making. We design evaluation protocols spanning cross-room, cross-user, and cross-dataset generalization settings using publicly available WiFi sensing benchmarks. Our study aims to establish a systematic framework for evaluating and improving the reliability of WiFi sensing systems deployed in previously unseen environments.

More broadly, we argue that future WiFi sensing research should complement accuracy reporting with calibration, uncertainty, and risk-aware evaluation metrics to enable trustworthy deployment in safety-critical scenarios.

***Index Terms*—WiFi Sensing, Human Activity Recognition, Conformal Prediction, Uncertainty Quantification, Domain Generalization, Out-of-Distribution Detection**

I. Introduction

Wireless sensing using commodity WiFi infrastructure has emerged as a promising alternative to camera-based and wearable sensing systems. By exploiting the interaction between radio frequency (RF) signals and human motion, WiFi-based sensing systems can infer a wide range of activities without requiring users to carry dedicated devices. Recent advances have demonstrated successful applications in gesture recognition, human activity recognition (HAR), occupancy monitoring, respiration tracking, fall detection, and smart healthcare environments. Compared to vision-based systems, WiFi sensing offers advantages in privacy preservation, lighting independence, and low deployment cost, making it attractive for real-world intelligent environments

A key enabler of modern WiFi sensing is Channel State Information (CSI), which provides fine-grained measurements of the wireless channel across multiple orthogonal frequency division multiplexing (OFDM) subcarriers. Human movement alters signal propagation paths, producing measurable changes in CSI that can be exploited for activity recognition. With the availability of CSI extraction tools and large-scale datasets, deep learning models have achieved impressive performance on benchmark WiFi sensing tasks. Convolutional neural networks, recurrent architectures, attention mechanisms, and more recently transformer-based models have significantly improved recognition accuracy under controlled evaluation settings.

Despite these advances, reliable deployment of WiFi sensing systems remains challenging. Wireless propagation is inherently environment-dependent. Changes in room geometry, furniture arrangement, wall materials, user orientation, hardware configuration, and multipath structure can substantially alter the observed CSI distribution. As a result, a model trained in one environment often experiences significant performance degradation when deployed in another. This phenomenon is commonly referred to as environment shift, domain shift, or distribution shift.

Most existing research addresses environment shift primarily through improved representation learning, domain adaptation, or domain generalization techniques. While these approaches often improve classification accuracy under unseen conditions, they do not directly address an equally important question:

When should the model trust its own prediction?

In real deployments, incorrect predictions are unavoidable. However, not all errors are equally problematic. A prediction made with low confidence naturally signals uncertainty and may trigger additional verification or human intervention. In contrast, a highly confident but incorrect prediction can lead to silent system failures that remain undetected. Such failures are particularly concerning in safety-sensitive applications including fall detection, elderly monitoring, healthcare assistance, and security monitoring, where incorrect decisions may carry significant consequences.

Modern deep neural networks are known to exhibit overconfidence, especially under distribution shift. Prior work in machine learning has shown that confidence scores derived from softmax probabilities often fail to accurately reflect predictive uncertainty. Consequently, models may assign extremely high confidence to incorrect predictions when encountering inputs that differ substantially from the training distribution. Although calibration and uncertainty estimation have been extensively studied in computer vision and natural language processing, their adoption within the WiFi sensing community remains limited.

This observation motivates a shift in perspective. Rather than evaluating WiFi sensing systems solely in terms of classification accuracy, we argue that reliability should be treated as a first-class objective. A practical sensing system should not only make accurate predictions but should also provide meaningful uncertainty estimates and indicate when its predictions should not be trusted.

To address this challenge, we investigate conformal prediction as a reliability layer for WiFi-based human activity recognition. Conformal prediction is a statistically grounded framework that converts point predictions into prediction sets while providing finite-sample coverage guarantees under appropriate assumptions. Unlike many uncertainty estimation techniques, conformal prediction operates as a post-hoc procedure and can be applied to existing classifiers without modifying their architecture or training process. These properties make it particularly attractive for deployment-oriented sensing applications.

Building on this idea, we propose **WiCaP (WiFi Conformal Activity Prediction)**, a framework designed to improve reliability under environment shift. WiCaP combines three complementary components. First, a **Domain-Contrastive Encoder (DCE)** learns activity representations that are less sensitive to environment-specific variations through supervised contrastive learning and adversarial environment decorrelation. Second, a **Conformal Calibration Layer (CCL)** transforms classifier outputs into uncertainty-aware prediction sets. Third, a **Selective Prediction Module (SPM)** leverages conformal uncertainty estimates to abstain on uncertain inputs rather than forcing potentially unreliable predictions.

In addition, we introduce the concept of **Silent Failure Rate (SFR)**, a reliability metric that measures the frequency of high-confidence incorrect predictions. While accuracy evaluates overall predictive performance, SFR focuses specifically on the most dangerous failure mode encountered during deployment: confident mistakes. We argue that SFR provides complementary information that is not captured by traditional evaluation metrics.

The objective of this work is not merely to improve recognition accuracy. Instead, we aim to establish a broader framework for evaluating and improving the reliability of WiFi sensing systems operating under realistic deployment conditions. To this end, we consider cross-room, cross-user, and cross-dataset evaluation protocols that reflect practical sources of distribution shift and assess performance using calibration, coverage, uncertainty, and abstention metrics in addition to standard accuracy.

The main contributions of this work are summarized as follows:

1) We identify and formalize the problem of silent failures in WiFi-based human activity recognition and introduce Silent Failure Rate (SFR) as a reliability-oriented evaluation metric.
2) We propose WiCaP, a reliability-aware framework that combines domain-invariant representation learning, conformal calibration, and selective prediction for WiFi sensing under environment shift.
3) We investigate the use of conformal prediction as a practical uncertainty quantification mechanism for WiFi sensing and analyze its behavior under cross-domain deployment scenarios.
4) We establish a comprehensive evaluation methodology that extends beyond classification accuracy to include calibration quality, coverage validity, uncertainty estimation, abstention behavior, and out-of-distribution robustness.
5) We advocate for reliability-aware evaluation protocols in WiFi sensing and argue that uncertainty reporting should accompany accuracy reporting in future deployment-oriented research.

The remainder of this paper is organized as follows. Section 2 reviews related work in WiFi sensing, domain generalization, uncertainty quantification, and conformal prediction. Section 3 formulates the problem and introduces reliability metrics. Section 4 presents the proposed WiCaP framework. Section 5 describes datasets, baselines, and evaluation protocols. Section 6 reports experimental results and analyses. Sections 7–10 discuss implications, limitations, future directions, and concluding remarks.

II. Related Work

A. 2.1 WiFi-Based Human Activity Recognition

WiFi-based human activity recognition (HAR) has become an active research area due to the widespread availability of wireless infrastructure and the ability of radio frequency signals to capture human motion without requiring wearable devices. Early WiFi sensing systems primarily relied on Received Signal Strength Indicator (RSSI) measurements and Doppler-based signal processing techniques for coarse-grained activity recognition. The release of Channel State Information (CSI) extraction tools by Halperin et al. enabled fine-grained sensing using commodity IEEE 802.11n devices, significantly expanding the capabilities of WiFi-based sensing systems.

Several pioneering systems demonstrated the feasibility of device-free activity recognition using wireless signals. WiSee introduced whole-home gesture recognition through Doppler-based wireless sensing, while CARM established a causal relationship between CSI dynamics and human movement characteristics. Subsequent work showed that CSI measurements could be used to recognize gestures, monitor respiration, estimate occupancy, and perform general human activity recognition with increasing accuracy.

The emergence of deep learning further accelerated progress in WiFi sensing. Convolutional neural networks (CNNs) have been used to extract spatial and frequency-domain features from CSI spectrograms, while recurrent neural networks (RNNs) and long short-term memory (LSTM) architectures model temporal dependencies in wireless signals. More recent approaches employ attention mechanisms, self-supervised representation learning, masked signal modeling, and transformer-based architectures to improve activity recognition performance. Large-scale benchmarks such as Widar 3.0 and NTU-Fi have facilitated systematic evaluation of deep learning approaches across users and environments.

Despite substantial progress, most existing WiFi HAR studies primarily evaluate performance using classification accuracy. Reliability, uncertainty estimation, and deployment-time confidence assessment remain relatively underexplored compared to accuracy optimization.

B. 2.2 Domain Adaptation and Generalization in WiFi Sensing

A major challenge in WiFi sensing is the sensitivity of CSI measurements to environmental conditions. Changes in room geometry, furniture arrangement, hardware placement, user orientation, and propagation characteristics often produce significant distribution shifts between training and deployment environments.

To address this issue, researchers have proposed domain adaptation and domain generalization techniques. Domain-Adversarial Neural Networks (DANN) learn environment-invariant representations through adversarial training using gradient reversal layers. Similar ideas have been adopted in WiFi sensing to reduce environment-specific information in learned feature representations.

Other approaches focus on transfer learning, meta-learning, and self-supervised adaptation. Widar 3.0 introduced a zero-effort cross-domain gesture recognition framework designed to improve robustness across users and environments. Subsequent work explored invariant feature learning, contrastive representation learning, and masked signal reconstruction objectives to improve generalization performance under distribution shift.

Although these methods often improve recognition accuracy in unseen environments, they typically produce deterministic predictions for all inputs. Consequently, a model may remain highly confident even when encountering samples that differ substantially from those observed during training. Existing domain adaptation approaches generally do not provide formal uncertainty estimates or mechanisms for abstaining on uncertain predictions.

C. 2.3 Uncertainty Quantification and Calibration in Deep Learning

Uncertainty quantification has become an important research area in modern machine learning, particularly for safety-critical applications. Neural networks often exhibit poor calibration, meaning that predicted confidence scores do not accurately reflect true correctness probabilities.

Bayesian Neural Networks model uncertainty through probability distributions over network parameters, but they are often computationally expensive and difficult to scale. Monte Carlo Dropout approximates Bayesian inference by performing multiple stochastic forward passes at inference time, while Deep Ensembles estimate predictive uncertainty using multiple independently trained models.

Post-hoc calibration methods provide a simpler alternative. Temperature Scaling has emerged as one of the most widely used calibration techniques due to its simplicity and effectiveness under in-distribution conditions. However, several studies have shown that calibration quality often deteriorates substantially under dataset shift, where confidence scores may become unreliable despite calibration on validation data.

While uncertainty estimation and calibration have been extensively studied in computer vision and natural language processing, their application to WiFi sensing remains relatively limited. Most WiFi HAR systems continue to rely on softmax confidence scores without evaluating calibration quality or deployment reliability.

D. 2.4 Conformal Prediction

Conformal prediction provides a statistically principled framework for uncertainty quantification. Originally introduced by Vovk et al., conformal prediction constructs prediction sets that satisfy finite-sample coverage guarantees under the assumption of exchangeability.

Unlike Bayesian methods, conformal prediction is model-agnostic and can be applied to any trained classifier. Recent advances have significantly improved the practicality of conformal methods for modern machine learning systems. Adaptive Prediction Sets (APS) and Regularized Adaptive Prediction Sets (RAPS) produce informative prediction sets while maintaining coverage guarantees. These methods have been successfully applied to image classification, medical imaging, protein structure prediction, language modeling, and time-series forecasting.

Several extensions have also been proposed for distribution-shifted settings. Weighted conformal prediction incorporates importance weighting to improve coverage under covariate shift, while other approaches investigate domain adaptation and risk-controlled prediction within the conformal framework.

Despite the rapid growth of conformal prediction research, its application to WiFi-based sensing remains largely unexplored. Existing WiFi sensing studies rarely evaluate coverage validity, prediction set quality, or conformal uncertainty under environmental distribution shift.

E. 2.5 Out-of-Distribution Detection and Selective Prediction

Out-of-distribution (OOD) detection seeks to identify inputs that differ significantly from the training distribution. Reliable OOD detection is particularly important in deployment scenarios where models may encounter previously unseen environments or activity patterns.

Maximum Softmax Probability (MSP) is one of the earliest and most widely used OOD detection baselines. Subsequent approaches introduced Mahalanobis-distance-based methods, energy-based scores, uncertainty estimation through ensembles, and density-based detectors. While these techniques often improve OOD detection performance, many rely on heuristic confidence measures whose behavior under distribution shift is difficult to characterize theoretically.

Selective prediction offers an alternative perspective. Rather than predicting on every input, a selective model may abstain when uncertainty exceeds a specified threshold. Geifman and El-Yaniv formalized selective classification as a trade-off between prediction coverage and prediction risk. More recent work has connected selective prediction with conformal risk control, enabling statistical guarantees on deployment-time risk.

The selective prediction paradigm is particularly relevant for WiFi sensing, where uncertain predictions may be more harmful than abstentions. However, systematic investigation of abstention mechanisms and reliability-aware deployment remains limited within the WiFi sensing literature.

F. 2.6 Summary and Research Gap

Existing WiFi sensing research has primarily focused on improving classification accuracy under environment shift through better architectures, representation learning techniques, and domain adaptation methods. At the same time, the machine learning community has developed a rich body of work on calibration, uncertainty quantification, conformal prediction, and selective prediction.

These two research directions have largely evolved independently. To the best of our knowledge, there has been no comprehensive study that systematically integrates domain-invariant representation learning, conformal uncertainty quantification, and selective prediction within a unified framework for WiFi-based human activity recognition.

This work aims to bridge that gap. Rather than focusing exclusively on accuracy, we investigate reliability-aware WiFi sensing through calibrated uncertainty estimation, prediction-set generation, and risk-aware abstention under environment shift. Our objective is not only to improve recognition performance but also to provide mechanisms that enable sensing systems to recognize and communicate uncertainty during deployment.

III. Problem Formulation

A. 3.1 WiFi CSI Signal Representation

Let

$$\mathcal{X} \subset \mathbb{R}^{T \times S \times F}$$

denote the space of Channel State Information (CSI) observations, where:

- T denotes the temporal window length,
- S denotes the number of antenna streams,
- F denotes the number of OFDM subcarriers.

A CSI sample $x \in \mathcal{X}$ is obtained from the channel frequency response $H(f, t)$ measured across time and frequency. Following standard WiFi sensing practice, both amplitude and phase information are used to construct activity representations.

Let

$$\mathcal{Y} = \{1, \ldots, K\}$$

denote the set of activity labels, where K is the number of activity classes.

The objective of WiFi-based human activity recognition is to learn a mapping

$$f : \mathcal{X} \to \mathcal{Y}$$

that predicts the underlying human activity from CSI measurements.

B. 3.2 Environment-Induced Distribution Shift

WiFi sensing is highly sensitive to environmental factors. Let $e \in \mathcal{E}$ represent an environment variable capturing:

- room geometry,
- furniture arrangement,
- wall materials,
- transmitter and receiver placement,
- hardware characteristics,
- human position and orientation.

The joint data distribution depends on the environment:

$$P^e(x, y)$$

Training data are collected from source environments e_s, while deployment occurs in a target environment e_t. As a result:

$$P^{e_s}(x) \neq P^{e_t}(x)$$

This distribution shift can significantly degrade recognition performance.

Unlike conventional vision tasks, WiFi sensing is strongly affected by multipath propagation. Even when the activity remains unchanged, its CSI representation may vary significantly across environments, leading to reduced model reliability during deployment.

C. 3.3 Reliability Beyond Accuracy

Most prior work evaluates performance using classification accuracy. However, accuracy alone does not capture prediction reliability.

Consider a classifier that outputs:

- predicted label $\hat{y}$,
- confidence score $c(x) \in [0, 1]$.

Two models may achieve identical accuracy while exhibiting very different confidence behaviors. In particular, highly confident incorrect predictions are more dangerous than low-confidence errors because they provide no indication of uncertainty.

This motivates the need for reliability-aware evaluation metrics.

D. 3.4 Silent Failure

We define a silent failure as a confident but incorrect prediction.

1) Definition 1 (Silent Failure): Given a classifier $f : \mathcal{X} \to \mathcal{Y}$ and a confidence threshold τ, a silent failure occurs for a sample (x, y) when:

$$f(x) \neq y \quad \text{and} \quad c(x) \geq \tau$$

Intuitively, the model makes an incorrect prediction with high confidence.

2) Definition 2 (Silent Failure Rate): For a dataset of N samples, the Silent Failure Rate (SFR) at confidence threshold τ is defined as:

$$\text{SFR}(\tau) = \frac{|\{i : f(x_i) \neq y_i \ \wedge \ c(x_i) \geq \tau\}|}{|\{i : c(x_i) \geq \tau\}|}. \tag{1}$$

SFR measures the proportion of highly confident predictions that are incorrect.

E. 3.5 Calibration Quality

Reliable deployment requires confidence scores to reflect prediction correctness.

A calibrated classifier should satisfy:

$$P(f(x) = y \mid c(x) = p) \approx p. \tag{2}$$

In other words, predictions assigned confidence p (e.g., 90%) should be correct approximately p of the time.

To quantify calibration quality, we use Expected Calibration Error (ECE).

1) Definition 3 (Expected Calibration Error): Let confidence scores be partitioned into M bins:

$$\{B_1, B_2, \ldots, B_M\}$$

The Expected Calibration Error is defined as:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\text{acc}(B_m) - \text{conf}(B_m)\right|. \tag{3}$$

where:

- $\text{acc}(B_m)$ is the empirical accuracy within bin m,
- $\text{conf}(B_m)$ is the average confidence within bin m.

Smaller ECE values indicate better calibration.

F. 3.6 Prediction Sets and Uncertainty

Traditional classifiers output a single label. In contrast, conformal prediction produces a set of plausible labels.

Let

$$\hat{\mathcal{C}}(x) \subseteq \mathcal{Y}$$

denote the prediction set for input x.

Examples:

$$\hat{\mathcal{C}}(x) = \{\text{Walking}\}$$

indicates high confidence, while

$$\hat{\mathcal{C}}(x) = \{\text{Walking}, \text{Running}\}$$

indicates ambiguity.

A large prediction set (e.g., $\hat{\mathcal{C}}(x) = \mathcal{Y}$) indicates high uncertainty.

Thus, prediction set size acts as a natural uncertainty signal.

G. 3.7 Selective Prediction

In deployment, it is often preferable to abstain rather than make unreliable predictions.

We define a selective prediction function:

$$g : \mathcal{X} \to \mathcal{Y} \cup \{\perp\}$$

where $\perp$ denotes abstention.

The model predicts a label when confidence is high and abstains otherwise.

This introduces a trade-off between:

- Coverage (fraction of predictions made),
- Accuracy (correctness of accepted predictions),
- Reliability (trustworthiness of predictions).

H. 3.8 Problem Statement

Given:

- A labeled source-domain dataset

$$\mathcal{D}_s = \{(x_i, y_i)\}_{i=1}^{N_s}$$

  collected from source environments,
- A calibration dataset $\mathcal{D}_c$,
- A target deployment environment exhibiting distribution shift,

our objective is to construct a prediction function that:

- achieves strong recognition performance,
- produces calibrated confidence estimates,
- generates informative prediction sets,
- minimizes silent failures,
- supports selective abstention.

Unlike conventional WiFi HAR systems that optimize only accuracy, the proposed framework explicitly targets reliability under distribution shift.

IV. Methodology

4.1 Framework Overview

WiCaP consists of three sequentially composed components:

1. Domain-Contrastive Encoder (DCE): A representation learning backbone that learns environment-invariant activity embeddings from labeled source-domain CSI data.

2. Weighted Conformal Calibration Layer (CCL): A post-hoc uncertainty calibration module that adapts prediction sets to a target environment using importance-weighted conformal prediction.

3. Selective Prediction Module (SPM): An abstention mechanism that converts conformal uncertainty into deployment-time prediction decisions.

Training occurs entirely on source-domain labeled data. During deployment, a small unlabeled sample from the target environment is used only for density-ratio estimation required by weighted conformal prediction. No retraining, fine-tuning, or gradient-based adaptation is performed after deployment.

This design separates representation learning from uncertainty calibration, allowing WiCaP to provide reliable uncertainty estimates without modifying the backbone architecture.

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4.2 Domain-Contrastive Encoder (DCE)

4.2.1 Motivation

The informativeness of conformal prediction sets depends strongly on the quality of the underlying feature representation. Poor representations produce ambiguous class probabilities and consequently large prediction sets.

The objective of DCE is therefore to learn representations that satisfy two properties:

1. Samples corresponding to the same activity but collected in different environments should be mapped to nearby regions of the embedding space.

2. Samples corresponding to different activities should remain well separated.

Unlike conventional supervised contrastive learning, WiCaP restricts positive pairs to samples sharing activity labels but originating from different environments. This pairing strategy explicitly suppresses environment-specific signal while preserving activity semantics.

Consequently, the encoder is optimized not merely for activity discrimination but for cross-environment activity consistency.

Assumption A1

The source training data contains samples collected from at least two distinct environments.

Environment identifiers are required only during training for contrastive pair construction and adversarial environment suppression. No environment labels are required during deployment.

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4.2.2 CSI Preprocessing

Raw CSI measurements are processed through the following pipeline:

• Phase sanitization using linear phase removal.

• Butterworth band-pass filtering with passband [0.1, 25] Hz.

• Per-sample z-score normalization.

• Short-Time Fourier Transform (STFT) using a Hann window.

The resulting representation is:

x R$^{(}T'F'S)$

where T' and F' denote STFT time and frequency bins and S denotes antenna streams.

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4.2.3 Encoder Architecture

The encoder

g : X $\rightarrow$ R^d

maps CSI observations into a d-dimensional embedding space.

Architecture:

• Stem: Two-layer CNN with BatchNorm and ReLU.

• Feature Extractor: Four residual blocks with channel sizes 32, 64, 128, 256.

• Temporal Aggregation: Lightweight self-attention over the temporal dimension.

• Projection Head: Two-layer MLP producing d = 128 dimensional embeddings.

The design intentionally avoids large transformer architectures to reduce computational cost and minimize overfitting to environment-specific structures.

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4.2.4 Contrastive Pre-training Objective

Let

(x$_i, y_i, e_i$)

denote a sample consisting of CSI observation, activity label, and environment label.

Positive pairs:

Samples with identical activity labels but different environments:

((x$_i, e_s), (x_j, e_t)$)

where

y$_i = y_j$

and

e$_s e_t$.

Negative pairs:

Samples belonging to different activities regardless of environment.

The normalized embedding is:

z = g(x) / ——g(x)——

The supervised contrastive loss is

L$_S upCon^(i) = (-1/|P(i)|)_{pP(i)} log[exp(sim(z_i, z_p)/)/_{aA(i)} exp(sim($

where P(i) denotes cross-environment positive samples and A(i) contains all remaining samples in the batch.

To further suppress environment-specific information, an auxiliary environment classifier

h : R^{d}ß$R^|E|$

is attached through a Gradient Reversal Layer (GRL).

The environment adversarial objective is

L$_e nv = -L_C E(h(GRL(g(x))), e)$

The full training objective is

$L_pretrain = L_SupCon + L_env + L_CE(y, y)$

where and are balancing hyperparameters.

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4.3 Weighted Conformal Calibration Layer (CCL)

4.3.1 Background

Given a trained classifier

f : X $\rightarrow$ ^{K}

and a nonconformity score function

s : X × Y $\rightarrow$ R

split conformal prediction constructs prediction sets with target coverage level

1 .

For calibration scores

s, ..., s_n

the conformal prediction set is

Ĉ(x) = y Y : s(x,y) q

where q is the calibration quantile.

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4.3.2 Nonconformity Score

We consider three candidate nonconformity scores.

Softmax Score:

$s_softmax(x, y) = 1 - p_y(x)$

APS Score:

$s_APS(x, y) =_{k:p_k(x)>p_y(x)} p_k(x) + up_y(x)$

where

u Uniform(0,1).

RAPS Score:

$s_RAPS(x, y) = s_APS(x, y) +_r egmax(o(x, y) - k_reg, 0)$

where o(x,y) denotes the class rank.

WiCaP adopts RAPS because it typically produces smaller prediction sets while maintaining empirical coverage.

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4.3.3 Weighted Conformal Prediction

The exchangeability assumption underlying standard conformal prediction is violated under environment shift.

To address this issue, WiCaP adopts the weighted conformal framework of Tibshirani et al. (2019).

Let

w(x) = $p_t(x)/p_s(x)$

denote the density ratio between target and source distributions.

Since these densities are unknown, a domain classifier

D : X $\rightarrow$ [0,1]

is trained to distinguish source samples from target samples.

The estimated importance weight is

ŵ(x) = D(x) / (1 - D(x))

Calibration scores are assigned corresponding weights, and the conformal threshold is obtained through a weighted empirical quantile.

This procedure reweights source calibration samples according to their similarity to the target environment.

When source and target distributions coincide, weighted conformal prediction reduces to standard conformal prediction.

Following Tibshirani et al. (2019), weighted conformal prediction asymptotically recovers target-domain coverage under covariate shift.

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4.3.4 Coverage Adaptation Property

Weighted conformal prediction does not generally preserve exact finite-sample guarantees under arbitrary covariate shift.

Instead, asymptotic target-domain coverage is recovered when density-ratio estimation is consistent.

Therefore, WiCaP inherits the theoretical properties of weighted conformal prediction rather than introducing a new coverage theorem.

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4.4 Selective Prediction Module (SPM)

4.4.1 Set Size as an Uncertainty Signal

The conformal prediction set

Ĉ(x)

naturally provides an uncertainty measure.

- —Ĉ(x)— = 1 indicates high confidence.
- Larger prediction sets indicate increased ambiguity.
- Very large prediction sets suggest substantial uncertainty.

Large conformal prediction sets are treated as uncertainty indicators rather than definitive evidence of out-of-distribution behavior.

Empirically, inputs originating from unseen environments tend to produce larger prediction sets due to increased ambiguity.

Consequently, prediction set size serves as a practical OOD proxy.

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4.4.2 Coverage–Accuracy Trade-off

The abstention threshold $_setcontrolsatrade-offbetween:$

1. Coverage Rate: Fraction of samples receiving predictions.
2. Accuracy on Accepted Inputs: Accuracy after abstained samples are removed.
3. Prediction Informativeness: Average conformal prediction set size.

Rather than reporting a single operating point, WiCaP evaluates the full coverage–accuracy frontier.

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4.4.3 Silent Failure Mitigation

Selective abstention reduces the frequency of high-confidence incorrect predictions by withholding predictions on uncertain inputs.

Rather than claiming formal finite-sample bounds on silent failure rate, WiCaP evaluates SFR empirically as a deployment-oriented reliability metric.

Future work may integrate conformal risk control methods to provide explicit SFR guarantees.

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4.5 Training and Deployment Pipeline

Phase 1 — Source-Domain Training

1. Collect labeled CSI data from multiple source environments.
2. Apply preprocessing described in Section 4.2.2.
3. Train DCE using $L_pretrain$.
4. Freeze encoder parameters.

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Phase 2 — Deployment Adaptation

1. Collect a small unlabeled sample from the target environment.

2. Train the density-ratio estimator D.

3. Estimate importance weights.

4. Compute weighted conformal thresholds using a labeled source calibration set.

5. Deploy WiCaP.

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Inference Procedure

For each input sample x:

1. Preprocess CSI observation.

2. Compute embedding z = g(x).

3. Obtain class probabilities.

4. Compute RAPS nonconformity scores.

5. Construct prediction set:

$\hat{C}$(x) = k : s(x,k) q_w

6. If

—$\hat{C}$(x)— $_set$

return "uncertain".

7. Else if

—$\hat{C}$(x)— = 1

return a hard prediction.

8. Otherwise return the conformal prediction set.

The deployment overhead consists only of prediction-set computation and threshold comparison, introducing negligible additional latency relative to the encoder forward pass.ON 4 HERE

V. Experimental Setup

5.1 Datasets

We evaluate WiCaP on three publicly available WiFi sensing benchmarks that represent different levels of distribution shift.

5.1.1 Widar 3.0

Widar 3.0 is a large-scale WiFi gesture recognition dataset collected using a USRP B210 software-defined radio operating at 2.4 GHz.

The dataset contains approximately 258,000 gesture instances spanning 22 gesture classes performed by 16 users across 5 indoor environments. CSI measurements are recorded from 30 OFDM subcarriers and multiple transmit-receive antenna pairs.

Widar 3.0 is selected because it provides explicit room-level domain variation, making it suitable for evaluating cross-environment generalization.

Cross-Room Protocol

Training environments:

R1, R2, R3

Test environments:

R4, R5

No samples from R4 or R5 are used during model training.

Cross-User Protocol

Leave-one-user-out evaluation is performed across all 16 users.

For each split, one user is held out for testing while the remaining users are used for training.

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5.1.2 UT-HAR

UT-HAR is a WiFi-based human activity recognition dataset containing seven activity classes:

• Walking
• Running
• Sitting
• Standing
• Lying Down
• Falling
• Picking Up Object

CSI measurements are collected using commodity IEEE 802.11n devices.

The dataset contains variability in user position, body orientation, and clothing conditions.

Cross-User Protocol

Leave-one-subject-out evaluation is used.

The dataset serves primarily as an external evaluation benchmark and transfer-learning target.

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5.1.3 NTU-Fi HAR

NTU-Fi HAR contains CSI recordings collected from 20 participants performing six human activities.

Measurements are obtained using commodity WiFi hardware with CSI extraction tools.

The dataset is widely used for benchmarking deep learning methods in WiFi sensing.

Cross-User Protocol

Leave-one-subject-out evaluation is performed across all participants.

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5.2 Evaluation Protocols

To evaluate robustness under progressively harder forms of distribution shift, we consider three protocols.

P1: Cross-Room Generalization

Training and testing occur on different physical environments within Widar 3.0.

This protocol evaluates robustness to changes in room geometry, furniture layout, and multipath structure.

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P2: Cross-User Generalization

Training and testing occur on disjoint users.

This protocol evaluates robustness to variations in body shape, movement style, and execution speed.

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P3: Cross-Dataset Transfer

Models are trained on Widar 3.0 and evaluated on UT-HAR or NTU-Fi HAR.

This represents the most challenging setting because the source and target domains differ simultaneously in:

• Hardware platform
• Data collection protocol
• Environment
• Participant population
• Activity distribution

A small unlabeled target-domain sample is used only for weighted conformal adaptation.

No target-domain labels are used for adaptation.

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5.3 Baseline Methods

We compare WiCaP against representative approaches from three categories.

Standard Supervised Learning

B1 — ERM

Standard empirical risk minimization using cross-entropy loss.

Represents conventional deployment without uncertainty handling.

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Domain Generalization and Adaptation

B2 — DANN

Domain-Adversarial Neural Network using gradient reversal training.

Represents adversarial domain-invariant learning.

B3 — Supervised Contrastive Learning (SupCon)

A supervised contrastive baseline without environment-specific pairing.

This baseline isolates the contribution of WiCaP's cross-environment contrastive objective.

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Uncertainty Quantification

B4 — Temperature Scaling

Post-hoc confidence calibration following Guo et al. (2017).

B5 — MC Dropout

Approximate Bayesian inference through stochastic dropout at inference time.

50 Monte Carlo samples are used.

B6 — Deep Ensemble

Ensemble of five independently trained ERM models.

Represents a strong uncertainty estimation baseline.

—

Proposed Method Ablations

B7 — DCE Only

Domain-Contrastive Encoder without conformal calibration.

B8 — Conformal Only

Weighted conformal prediction applied on top of ERM.

B9 — WiCaP without Importance Weighting

Standard conformal prediction without density-ratio adaptation.

B10 — Full WiCaP

Complete proposed framework.

—

5.4 Evaluation Metrics

We intentionally evaluate reliability in addition to accuracy.

M1 — Accuracy (ACC)

Standard top-1 classification accuracy.

For selective prediction methods, accuracy is computed only on accepted predictions.

—

M2 — Coverage Rate (CVG)

Fraction of samples receiving a prediction rather than abstention.

CVG = Number of Accepted Samples / Total Samples

—

M3 — Empirical Coverage (EC)

Fraction of test samples whose true label is contained within the conformal prediction set.

EC = P(y Ĉ(x))

This metric evaluates whether empirical coverage matches the target level (1).

—

M4 — Average Set Size (ASS)

Average cardinality of conformal prediction sets.

ASS = E[—Ĉ(x)—]

Smaller values indicate more informative uncertainty estimates.

—

M5 — Expected Calibration Error (ECE)

Confidence calibration measured using 15 equal-width bins.

Lower values indicate better calibration.

—

M6 — Silent Failure Rate (SFR)

Silent Failure Rate is computed using confidence threshold = 0.9.

SFR(0.9) measures the fraction of highly confident predictions that are incorrect.

This metric is particularly important because silent failures represent the most dangerous deployment scenario.

—

M7 — AUROC for OOD Detection

Area Under the Receiver Operating Characteristic curve.

Prediction set size is used as the uncertainty score.

This metric evaluates the ability of WiCaP to identify inputs originating from unseen environments.

—

5.5 Statistical Testing

All experiments are repeated using five independent random seeds.

Results are reported as:

Mean ± Standard Deviation.

For pairwise comparisons between methods, statistical significance is evaluated using a paired t-test at significance level:

p ¡ 0.05

This prevents conclusions from being drawn based on random initialization effects.

—

5.6 Implementation Details

All models are implemented in PyTorch.

Encoder

Embedding dimension:

d = 128

—

Optimization

Optimizer:

AdamW

Learning rate:

TABLE I
ERM PERFORMANCE

Metric	Value
Accuracy	98.40%
SFR (Selective Failure Rate)	1.60%

$3 \times 10^{-}4$
Weight decay:
$1 \times 10^{-}4$
Batch size:
256
Training epochs:
200
—
Contrastive Learning
Temperature parameter:
= 0.07
—
Data Augmentation
During training, the following augmentations are applied:
• Time-frequency masking
• Gaussian noise injection
• Random subcarrier dropout
—
Weighted Conformal Prediction
Default coverage level:
1 = 0.90
Calibration size:
n = 100
unless otherwise stated.
—
Density Ratio Estimation
The domain classifier used for density-ratio estimation is a three-layer MLP trained using binary cross-entropy.
—
Compute Resources
Experiments are conducted on a single NVIDIA A100 or V100 GPU.
Pretraining is expected to require approximately 4–6 hours on Widar 3.0.
Conformal calibration requires less than one minute and is performed once during deployment.

VI. EXPERIMENTAL RESULTS

A. Baseline Performance (ERM)

The Empirical Risk Minimization (ERM) model serves as the baseline with no abstention mechanism. It achieves strong classification performance but lacks uncertainty awareness.

Although ERM achieves high accuracy, it makes overconfident incorrect predictions, motivating the need for uncertainty-aware methods.

TABLE II
APS RESULTS

Method	Coverage	ASS	Gap	q
APS @ 90%	99.20%	1.0220	9.20%	0.8966
APS @ 95%	99.60%	1.0360	4.60%	0.9483

TABLE III
RAPS RESULTS

Method	Coverage	ASS	Gap	q	λ
RAPS @ 90%	99.20%	1.0220	9.20%	0.8966	0.001
RAPS @ 95%	99.40%	1.0300	4.40%	0.9507	0.050

B. Adaptive Prediction Sets (APS)

APS provides probabilistic prediction sets with formal coverage guarantees.

APS achieves strong coverage guarantees but slightly increases the average set size (ASS), reflecting the trade-off between reliability and prediction specificity.

C. Regularized APS (RAPS)

RAPS introduces a regularization term to reduce prediction set size while maintaining coverage.

RAPS provides a measurable improvement at the 95% level, reducing ASS and coverage gap compared to APS, while maintaining similar performance at 90%.

D. APS vs RAPS Comparison

The comparison highlights that RAPS achieves slightly tighter prediction sets at higher confidence levels.

E. Selective Prediction Analysis

Selective prediction allows the model to abstain on uncertain samples using a confidence threshold τ.

A key observation is that the Selective Failure Rate (SFR) plateaus at 0.60% between $\tau = 0.80$ and $\tau = 0.90$. This indicates that certain confidently incorrect predictions persist despite increasing the threshold.

This behavior highlights a fundamental limitation of heuristic threshold-based abstention methods and motivates the use of conformal prediction, which provides statistically valid guarantees.

F. Key Insights

- ERM achieves high accuracy but lacks reliability under uncertainty.
- APS ensures strong coverage but increases prediction set size.
- RAPS improves efficiency by reducing set size while maintaining coverage.
- Selective prediction reduces error but fails to eliminate overconfident mistakes.
- Conformal methods provide a principled solution to uncertainty quantification.

—

TABLE IV
APS VS RAPS COMPARISON

Method	Coverage	ASS
APS @ 90%	99.20%	1.0220
RAPS @ 90%	99.20%	1.0220
APS @ 95%	99.60%	1.0360
RAPS @ 95%	99.40%	1.0300

TABLE V
SELECTIVE PREDICTION RESULTS

Threshold (τ)	Accuracy	Coverage	Abstention	SFR
ERM (no abstention)	98.40%	100.00%	0.00%	1.60%
0.70	98.7928%	99.40%	0.60%	1.20%
0.80	99.3902%	98.40%	1.60%	0.60%
0.90	99.3865%	97.80%	2.20%	0.60%
0.95	99.5876%	97.00%	3.00%	0.40%

VII. DISCUSSION

7.1 Beyond Accuracy as the Sole Evaluation Metric

A central motivation of this work is that accuracy alone may be insufficient for evaluating WiFi sensing systems intended for real-world deployment.

Traditional WiFi sensing research has largely focused on maximizing classification accuracy under increasingly challenging domain shifts. While accuracy remains an important measure of predictive performance, it does not fully characterize model reliability.

In particular, accuracy does not distinguish between:

• Low-confidence errors

• High-confidence errors

A model that makes mistakes while expressing uncertainty may be preferable to a model that makes the same number of mistakes while being highly confident.

From a deployment perspective, the latter failure mode is often more problematic because it can mislead downstream decision-makers into trusting incorrect predictions.

This observation motivates the introduction of Silent Failure Rate (SFR), which explicitly measures the frequency of confident incorrect predictions.

Rather than replacing accuracy, we argue that future WiFi sensing evaluations should complement accuracy with additional reliability metrics such as:

• Expected Calibration Error (ECE)

• Silent Failure Rate (SFR)

• Empirical Coverage

• Coverage–Accuracy Trade-off Curves

Together, these metrics provide a more complete characterization of deployment reliability.

—

7.2 Conformal Prediction as a Reliability Layer

Conformal prediction should not be viewed as a replacement for representation learning, domain adaptation, or domain generalization.

Instead, it provides a complementary uncertainty quantification layer that can be applied to an existing classifier.

The effectiveness of conformal prediction depends strongly on the quality of the underlying model.

If the learned representation is highly discriminative, conformal prediction sets tend to be small and informative.

Conversely, if the representation is weak, conformal prediction sets may become excessively large despite maintaining coverage.

This observation motivates the joint design of WiCaP:

1. The Domain-Contrastive Encoder aims to improve representation quality under environment shift.

2. Weighted conformal prediction aims to provide uncertainty estimates that remain meaningful under distribution shift.

3. Selective prediction converts uncertainty information into deployment-time decisions.

Viewed from this perspective, conformal prediction and representation learning address complementary aspects of reliability.

—

7.3 Practical Deployment Considerations

A practical challenge in uncertainty-aware WiFi sensing is balancing theoretical guarantees with deployment cost.

WiCaP assumes access to a small unlabeled sample collected from the target environment before deployment.

Such samples are used only for density-ratio estimation and do not require manual annotation.

Compared with approaches that require labeled target-domain data or environment-specific fine-tuning, this requirement is relatively lightweight.

Nevertheless, the feasibility of collecting deployment-time samples depends on the application scenario.

For example:

• Smart homes may permit short calibration procedures.

• Emergency-response applications may not.

Understanding this trade-off is important when evaluating the practical applicability of uncertainty-aware sensing systems.

—

7.4 Relationship to Domain Generalization

Domain generalization methods attempt to learn representations that remain robust across unseen environments.

WiCaP shares this objective through its Domain-Contrastive Encoder.

However, even a perfectly generalized representation cannot guarantee that uncertainty estimates remain calibrated under future distribution shifts.

Conversely, uncertainty calibration alone cannot compensate for poor representations.

The proposed framework therefore combines both perspectives:

• Representation learning improves robustness.

• Conformal calibration improves reliability.

This combination reflects the broader view that trustworthy sensing systems require both accurate representations and meaningful uncertainty estimates.

—

7.5 Limitations

Despite its potential advantages, WiCaP has several limitations.

L1 — Marginal Coverage Rather Than Conditional Coverage

Conformal prediction provides guarantees averaged over the data distribution.

Coverage is not guaranteed independently for every activity class.

As a result, some classes may receive lower empirical coverage than others.

Developing class-conditional coverage guarantees remains an active research area.

—

L2 — Dependence on Calibration Data

Weighted conformal prediction requires calibration information.

Although WiCaP only requires unlabeled target-domain samples for density-ratio estimation, performance may degrade when target-domain data are extremely limited.

Future work may investigate zero-shot or calibration-free alternatives.

—

L3 — Hardware Heterogeneity

CSI measurements are affected by hardware-specific processing pipelines, including:

- Automatic Gain Control (AGC)
- Phase distortions
- Sampling-rate differences

Consequently, cross-hardware transfer remains challenging.

The datasets considered in this work provide only partial coverage of real-world hardware diversity.

—

L4 — Open-Set Activities

WiCaP assumes that deployment activities belong to the same label space used during training.

When previously unseen activities occur, large prediction sets may signal uncertainty.

However, the framework does not explicitly solve open-set recognition.

Developing conformal methods for open-world activity recognition remains an important direction for future research.

—

L5 — Non-Stationary Environments

The current framework assumes that the target environment remains approximately stationary after calibration.

In practice, environmental conditions may change due to:

- Furniture movement
- Occupancy changes
- Hardware repositioning

Such changes may gradually invalidate calibration assumptions.

Online conformal recalibration is a promising direction for addressing this challenge.

—

7.6 Broader Implications

The broader contribution of this work is not limited to WiFi sensing.

Many sensing systems—including radar sensing, acoustic sensing, wearable sensing, and vision-based activity recognition—face similar challenges:

- Distribution shift
- Calibration failure
- Overconfident predictions

The methodology explored in WiCaP suggests a general framework in which uncertainty quantification is treated as a first-class design objective rather than an afterthought.

As machine learning systems continue to move into safety-critical and human-centered environments, the ability to communicate uncertainty may become as important as predictive accuracy itself.

VIII. Research Contribution and Scope

The primary contribution of this work is not the invention of a new uncertainty quantification theory, but rather the introduction of a reliability-centered framework for WiFi sensing under environment shift.

Existing WiFi human activity recognition research has largely focused on improving classification accuracy under increasingly challenging domain-shift scenarios. While significant progress has been made in domain adaptation, domain generalization, and self-supervised representation learning, comparatively little attention has been devoted to the reliability of predictions produced after deployment. In particular, current evaluation protocols rarely investigate whether model confidence remains trustworthy when sensing conditions differ from those observed during training.

This work addresses that gap through three contributions.

C1 — Reliability-Oriented Problem Formulation

We formalize the silent failure problem in WiFi sensing: situations in which a model produces an incorrect prediction with high confidence under environment shift. While miscalibration under distribution shift has been extensively studied in machine learning, its implications for WiFi sensing systems have not been systematically characterized.

To quantify this behavior, we introduce Silent Failure Rate (SFR), a metric that measures the proportion of highly confident predictions that are incorrect. Unlike overall accuracy, SFR directly captures a deployment-relevant failure mode that is particularly important in safety-sensitive applications such as fall detection, occupancy monitoring, and elder-care systems.

C2 — Conformal Reliability Layer for WiFi Sensing

We introduce WiCaP, a framework that combines WiFi sensing models with conformal prediction to obtain statistically valid uncertainty estimates under deployment-time distribution shift.

Rather than forcing a single prediction for every input, WiCaP produces prediction sets with finite-sample coverage guarantees. The framework additionally incorporates weighted conformal prediction to better account for discrepancies between source and target environments.

To the best of our knowledge, conformal prediction has not been systematically studied within the WiFi HAR literature, despite its attractive theoretical guarantees and minimal deployment overhead.

C3 — Representation Learning for Informative Prediction Sets

Conformal methods provide coverage guarantees independently of the underlying model; however, the practical usefulness of prediction sets depends heavily on representation quality.

To improve prediction set informativeness, we propose the Domain-Contrastive Encoder (DCE), which encourages activity representations to be invariant across environments while remaining discriminative across activity classes.

The proposed encoder is specifically designed to improve conformal prediction efficiency by reducing average prediction set size while preserving coverage validity.

8.1 Scope of Contributions

The contributions of this work should be interpreted within an applied machine learning context.

WiCaP does not introduce a fundamentally new conformal prediction algorithm. Instead, the contribution lies in demonstrating how reliability guarantees can be integrated into WiFi sensing systems through the combination of:

• environment-aware representation learning,
• weighted conformal calibration, and
• selective prediction.

The resulting framework provides a practical pathway toward reliability-aware deployment of WiFi sensing models without requiring retraining or extensive target-domain labeling.

8.2 Relationship to Existing Literature

This work draws inspiration from three research communities:

• WiFi sensing and human activity recognition,
• uncertainty quantification and calibration,
• conformal prediction and risk control.

Our goal is to bridge these communities rather than advance only one of them in isolation.

From the WiFi sensing perspective, the key contribution is the introduction of reliability evaluation alongside accuracy.

From the conformal prediction perspective, the contribution is the application and adaptation of coverage-based uncertainty quantification to CSI-based sensing under environment shift.

From the deployment perspective, the contribution is a practical framework that can be applied to existing WiFi sensing architectures with minimal modification.

8.3 Reproducibility and Scientific Integrity

All methodological claims in this paper are independently verifiable through experiments on publicly available datasets.

Theoretical claims are restricted to guarantees supported by existing conformal prediction theory and weighted conformal prediction literature.

Empirical claims are evaluated through controlled experiments under cross-room, cross-user, and cross-dataset protocols.

We intentionally separate theoretical guarantees from empirical observations and avoid making claims that are not supported by either analysis or experimental evidence.

This distinction is particularly important because coverage guarantees and predictive accuracy measure different properties of a deployed system. A method may achieve valid coverage while exhibiting poor accuracy, or vice versa. Consequently, both reliability and predictive performance must be evaluated jointly.

IX. Future Work

While WiCaP demonstrates how reliability guarantees can be integrated into WiFi-based human activity recognition systems, several important research challenges remain open. We outline promising directions for future work.

9.1 Conditional Coverage Guarantees

The conformal prediction framework employed in WiCaP provides marginal coverage guarantees averaged across the test distribution. However, practical deployments may require stronger guarantees for specific activity classes.

For example, in fall detection applications, ensuring high coverage for rare but safety-critical activities may be more important than achieving uniform average coverage across all classes.

Future research could explore:

• class-conditional conformal prediction,
• group-conditional coverage,
• activity-specific risk guarantees,
• localized conformal prediction methods.

Such approaches could provide stronger reliability guarantees for high-risk sensing tasks while maintaining practical deployment feasibility.

9.2 Online and Continual Conformal Recalibration

WiCaP assumes that the deployment environment remains relatively stable after calibration. In real-world settings, however, wireless propagation characteristics may evolve over time due to:

• furniture rearrangement,
• occupancy changes,
• environmental dynamics,
• hardware aging.

An important extension is online conformal prediction, where calibration statistics are continuously updated using recently observed data.

Developing theoretically grounded online calibration procedures for non-stationary wireless environments represents a promising direction for reliable long-term deployment.

9.3 Open-World Activity Recognition

Current WiCaP experiments assume that all activities encountered during deployment belong to the training label space.

Real-world deployments may violate this assumption. Users can perform activities that were never observed during training, producing signals that do not correspond to any known class.

Although conformal prediction naturally tends to produce larger prediction sets on such inputs, a formal treatment of unknown activities remains an open problem.

Future work could combine:

• open-set recognition,

• novelty detection,

• conformal uncertainty estimation,

to develop WiFi sensing systems capable of identifying genuinely unseen activities.

9.4 Hardware-Invariant Representation Learning

One of the most significant practical challenges in WiFi sensing is hardware variability.

CSI measurements collected using different chipsets often exhibit systematic differences due to:

• antenna characteristics,

• firmware processing,

• phase distortion,

• automatic gain control behavior.

Current domain generalization approaches primarily focus on environmental variation rather than hardware variation.

Future research may investigate hardware-invariant pre-training objectives that explicitly remove device-specific information from learned representations.

Such methods could substantially reduce the need for deployment-time calibration and improve cross-platform interoperability.

9.5 Multi-Modal Reliability-Aware Sensing

Modern sensing systems increasingly combine multiple sensing modalities, including:

• WiFi CSI,

• inertial sensors,

• microphones,

• cameras,

• radar.

Each modality provides complementary information and may fail under different conditions.

Extending conformal prediction to multi-modal sensing systems would enable prediction sets that account for uncertainty across multiple information sources simultaneously.

A particularly interesting direction is modality-aware conformal fusion, where uncertainty estimates adapt dynamically depending on the reliability of each sensor stream.

9.6 Active Deployment and Data Collection

The abstention mechanism in WiCaP provides a natural signal indicating where the model lacks confidence.

Rather than treating abstentions as failures, future systems could use them as triggers for targeted data acquisition.

For example, when a deployment environment consistently produces uncertain predictions, the system could request additional calibration data specifically from those regions of the feature space.

This would create a closed-loop framework combining:

• uncertainty estimation,

• active learning,

• domain adaptation,

• continual calibration.

Such systems could improve reliability over time while minimizing labeling effort.

9.7 Reliability-Aware Benchmarking Standards

Current WiFi sensing benchmarks primarily emphasize classification accuracy.

As sensing systems move toward real-world deployment, evaluation protocols should expand to include reliability metrics such as:

• Expected Calibration Error (ECE),

• Silent Failure Rate (SFR),

• empirical coverage validity,

• abstention-performance trade-offs.

Future benchmark design should encourage reporting both predictive performance and reliability characteristics.

Establishing community-wide reliability evaluation standards would improve comparability across methods and promote the development of safer sensing systems.

9.8 Formal Safety Guarantees for Sensing Applications

Many envisioned applications of WiFi sensing involve safety-critical decisions, including:

• fall detection,

• healthcare monitoring,

• assisted living,

• occupancy-based automation.

Future work could extend recent advances in conformal risk control to provide guarantees on task-specific safety metrics.

Examples include:

• bounded false-negative rates,

• bounded silent-failure rates,

• minimum detection sensitivity,

• application-specific risk constraints.

Such guarantees would move WiFi sensing beyond accuracy-oriented evaluation toward formally verifiable reliability objectives.

9.9 Foundation Models for Wireless Sensing

Recent developments in large-scale self-supervised learning suggest the possibility of foundation models for wireless sensing.

Training on massive unlabeled CSI corpora collected across diverse environments could yield representations that generalize far beyond current task-specific models.

An interesting future direction is the integration of foundation-model pre-training with conformal uncertainty estimation.

In this setting, large-scale pre-training would provide strong representations, while conformal calibration would provide deployment-time reliability guarantees.

The combination may offer a scalable path toward robust and trustworthy wireless sensing systems across diverse environments and hardware platforms.

X. Conclusion

WiFi-based human activity recognition has achieved remarkable progress over the past decade, with recent advances in deep learning substantially improving performance across a wide range of sensing tasks. Despite these improvements, the majority of existing research continues to evaluate systems primarily through classification accuracy, even though real-world deployments inevitably encounter distribution shifts caused by changes in environments, users, hardware platforms, and sensing conditions.

This paper argues that accuracy alone is insufficient for evaluating deployable WiFi sensing systems. A model that achieves high accuracy while remaining poorly calibrated may produce confident but incorrect predictions under distribution shift, creating failure modes that are difficult to detect and potentially harmful in safety-sensitive applications. Reliable deployment therefore requires not only accurate predictions, but also trustworthy uncertainty estimates and principled mechanisms for identifying situations in which the model should refrain from making a decision.

To address this challenge, we presented WiCaP, a reliability-oriented framework for WiFi-based human activity recognition under environment shift. WiCaP combines three complementary components:

- a Domain-Contrastive Encoder (DCE) that learns environment-invariant activity representations,
- a Weighted Conformal Calibration Layer (CCL) that provides statistically valid uncertainty quantification under domain shift,
- and a Selective Prediction Module (SPM) that converts uncertainty estimates into principled abstention decisions.

Unlike conventional confidence-based approaches, WiCaP produces prediction sets with formal finite-sample coverage guarantees and explicitly identifies uncertain inputs rather than forcing potentially unreliable predictions. The framework requires only a small unlabeled calibration set from the deployment environment and introduces negligible computational overhead after training.

Beyond the proposed methodology, this work introduces a broader reliability perspective for WiFi sensing research. We formalized the silent failure problem, proposed Silent Failure Rate (SFR) as a deployment-oriented evaluation metric, and argued for the inclusion of calibration and coverage measures alongside traditional accuracy metrics. These additions provide a more complete assessment of model behavior under realistic deployment conditions.

More broadly, the central message of this work is that future wireless sensing systems should be evaluated not only by how often they are correct, but also by how reliably they recognize uncertainty. As WiFi sensing continues to expand into domains such as healthcare monitoring, elder care, smart environments, and safety-critical decision support, reliability-aware evaluation will become increasingly important. Conformal prediction provides one promising pathway toward this goal by offering rigorous uncertainty guarantees while remaining compatible with existing sensing architectures.

We hope that this work encourages a shift toward reliability-centered wireless sensing research and serves as a foundation for future studies at the intersection of WiFi sensing, uncertainty quantification, and trustworthy machine learning.

APPENDIX A
THEORETICAL ANALYSIS

A.1 Coverage Properties of Weighted Conformal Prediction

The reliability guarantees of WiCaP are inherited from the weighted conformal prediction framework used in the calibration layer.

Let

D$_c = (x_i, y_i)_{i=1}^n$

denote a labeled calibration set and let

s(x,y)

be the chosen nonconformity score function.

For a target miscoverage level (0,1), weighted conformal prediction computes a weighted empirical quantile q$_w andconstructsthepredictionset$

Ĉ$_w(x) = yY : s(x,y)q_w.$

Unlike standard split conformal prediction, calibration and test samples are not assumed to be identically distributed. Instead, weighted conformal prediction accounts for covariate shift through importance weights

w(x) = p$_t(x)/p_s(x),$

where p$_s andp_t denotethesourceandtargetfeaturedistributionsrespe$

Under standard covariate-shift assumptions and consistent estimation of the density ratio, weighted conformal prediction provides approximate marginal coverage over the target distribution.

Theorem A1 (Weighted Conformal Coverage)

Assume:

1. Calibration and test samples satisfy a covariate-shift model,

2. The conditional label distribution is invariant,

P$_t(y|x) = P_s(y|x),$

3. Importance weights are estimated consistently.

Then the weighted conformal prediction set Ĉ$_w satisfiesasymptotictarget-domaincoverage:$

P$_t(Y\hat{C}_w(X))$ß1

as the calibration sample size n tends to infinity.

This result follows from the weighted conformal prediction framework of Tibshirani et al. (2019) and justifies the use of weighted calibration when source and target environments differ.

Importantly, the guarantee applies to marginal coverage under the target distribution rather than pointwise coverage for individual inputs.

A.2 Prediction Set Informativeness

Coverage alone does not determine whether conformal prediction is useful in practice.

A trivial predictor that always returns the full label set achieves perfect coverage but provides no actionable information.

Consequently, an important objective is to minimize prediction set size while maintaining valid coverage.

Definition A1 (Prediction Set Efficiency)

For a conformal predictor Ĉ, define the efficiency metric

ASS = E[—Ĉ(X)—],

where ASS denotes Average Set Size.

Smaller values indicate more informative prediction sets.

A.3 Representation Quality and Prediction Set Size

The practical usefulness of conformal prediction depends heavily on the quality of the underlying classifier.

Let

p$_{(}1)(x)$

and

p$_{(}2)(x)$

denote the largest and second-largest predicted class probabilities.

Define the confidence gap

gap(x) = p$_{(}1)(x)p_{(}2)(x)$.

A larger gap indicates stronger separation between the most likely class and competing alternatives.

Proposition A1

For ranking-based conformal methods such as APS and RAPS, the expected prediction set size decreases as the expected confidence gap increases.

Intuitively, when the classifier strongly prefers one class over all others, fewer labels satisfy the conformal inclusion criterion, resulting in smaller prediction sets.

Conversely, uncertain predictions produce flatter probability distributions and therefore larger prediction sets.

This observation motivates the use of the Domain-Contrastive Encoder.

By encouraging activity-discriminative and environment-invariant representations, DCE improves class separation in the embedding space and consequently increases the probability gap between competing classes.

Although the exact relationship depends on the classifier architecture and calibration quantile, improved representation quality generally translates into more informative conformal prediction sets.

A.4 Selective Prediction and Silent Failure Reduction

WiCaP uses conformal prediction set size as an abstention signal.

Let

$_{s}et$

denote the abstention threshold.

The selective predictor is defined as

f$_{s}el(x) = prediction, if|\hat{C}(x)| <_{s} etabstain, otherwise.$

The accepted-risk framework from selective prediction theory suggests that abstaining on highly uncertain inputs can improve accuracy among accepted predictions.

In the context of WiFi sensing, this mechanism specifically targets silent failures:

• confident incorrect predictions,

• predictions made under severe domain shift,

• ambiguous CSI observations.

Therefore, prediction-set size acts simultaneously as

(i) an uncertainty estimate,

(ii) an OOD indicator,

and

(iii) a selective prediction criterion.

A.5 Discussion of Theoretical Guarantees

The guarantees provided by WiCaP should be interpreted carefully.

First, conformal prediction provides marginal coverage guarantees rather than class-conditional guarantees.

Second, weighted conformal prediction relies on the covariate-shift assumption and the quality of density-ratio estimation.

Third, coverage guarantees do not imply high accuracy. A model may achieve valid coverage while still producing large prediction sets.

For this reason, WiCaP evaluates reliability through multiple complementary metrics:

• Coverage Validity (CCV),

• Average Set Size (ASS),

• Expected Calibration Error (ECE),

• Silent Failure Rate (SFR),

rather than relying on any single metric.

Taken together, these metrics provide a more complete characterization of deployment-time reliability under environment shift.

APPENDIX B
DATASET STATISTICS AND PREPROCESSING DETAILS

This appendix provides detailed information regarding dataset characteristics, preprocessing procedures, feature extraction, and evaluation splits used throughout the experiments.

B.1 Dataset Summary

Table B1 summarizes the datasets used in this work.

Table B1: Dataset Characteristics

Dataset — Activities — Participants — Environments — Hardware ———————————————— Widar 3.0 — 22 — 16 — 5 Rooms — USRP B210 SDR UT-HAR — 7 — 6 — 1 Room — 802.11n Router NTU-Fi HAR — 6 — 20 — Multiple — Intel 5300 NIC

These datasets collectively provide diversity in:

• activity classes, • sensing environments, • participant populations, • WiFi hardware platforms.

This diversity enables evaluation under realistic deployment-time distribution shifts.

B.2 Widar 3.0 Processing Pipeline

B.2.1 Raw CSI Structure

Each Widar sample consists of CSI measurements collected from:

• 3 transmit antennas, • 3 receive antennas, • 30 OFDM subcarriers.

The resulting CSI tensor contains:

T = 2000 temporal samples, S = 9 antenna-pair streams, F = 30 subcarriers.

Raw tensor shape:

(2000 × 9 × 30)

B.2.2 Environment Split

Cross-room evaluation follows the protocol:

Training Rooms: R1, R2, R3

Testing Rooms: R4, R5

This protocol evaluates generalization to previously unseen physical environments.

B.2.3 User Split

For cross-user evaluation, leave-one-user-out validation is used.

At each fold:

• 15 users are used for training, • 1 user is held out for testing.

Results are averaged across all folds.

B.3 UT-HAR Processing Pipeline

UT-HAR contains CSI recordings for seven activities:

1. Walking 2. Running 3. Sitting 4. Standing 5. Lying Down 6. Falling 7. Picking Up Object

Raw tensor dimensions:

T = 250 S = 3 F = 30

Tensor shape:

(250 × 3 × 30)

Cross-user evaluation is performed using leave-one-subject-out validation.

B.4 NTU-Fi HAR Processing Pipeline

The NTU-Fi HAR subset contains six activity categories collected from twenty participants.

Raw tensor dimensions:

T = 500 S = 3 F = 30

Tensor shape:

(500 × 3 × 30)

Evaluation follows the official participant-based split protocol whenever available.

B.5 CSI Preprocessing

Raw CSI measurements contain several sources of noise including:

• hardware-induced phase offsets, • packet timing jitter, • static background reflections, • high-frequency interference.

To improve signal quality, the following preprocessing pipeline is applied to all datasets.

B.5.1 Phase Sanitization

Raw CSI phase measurements are corrected using standard linear phase removal techniques.

This procedure reduces:

• carrier frequency offset, • sampling frequency offset, • hardware phase distortion.

Following phase sanitization, phase values become more consistent across packets.

B.5.2 Band-Pass Filtering

A Butterworth band-pass filter is applied:

Passband: 0.1 Hz – 25 Hz

The lower cutoff suppresses static environmental reflections.

The upper cutoff removes high-frequency noise unrelated to human motion.

B.5.3 Amplitude Normalization

For each CSI sample, amplitude values are standardized using z-score normalization:

x$_norm = (x)/$

where and are computed over the sample.

This step reduces sensitivity to absolute signal strength variations.

B.5.4 Outlier Handling

Packets containing corrupted CSI measurements are removed using standard amplitude thresholding procedures.

Missing values are linearly interpolated when necessary.

B.6 Time–Frequency Feature Extraction

Human activities generate characteristic Doppler and micro-Doppler signatures.

To capture these patterns, Short-Time Fourier Transform (STFT) is applied.

Parameters:

• Hann window • 50• Fixed window length across datasets

For each antenna stream, STFT produces a time-frequency representation.

The resulting tensor is stacked across antenna streams and provided as input to the encoder.

B.6.1 Widar Representation

After STFT:

T = 100 F = 20 S = 9

Final tensor shape:

(100 × 20 × 9)

B.6.2 UT-HAR Representation

After STFT:

T and F are chosen to maintain consistent frequency resolution.

Final dimensions are adjusted to match encoder requirements.

B.6.3 NTU-Fi Representation

The same preprocessing and feature extraction pipeline is applied.

This ensures that differences in performance reflect domain shift rather than preprocessing inconsistencies.

B.7 Data Augmentation

To improve robustness during training, the following augmentations are applied.

B.7.1 Gaussian Noise Injection

Additive Gaussian noise:

N(0, 0.01^2)

is applied to CSI amplitude values.

B.7.2 Time-Frequency Masking

Random contiguous regions in the spectrogram are masked.

This augmentation encourages robustness to partial signal corruption.

B.7.3 Subcarrier Dropout

Random subsets of subcarriers are removed during training.

This simulates packet loss and hardware variability.

B.8 Calibration Set Construction

For deployment-time conformal calibration, a small calibration set is collected from the target domain.

Calibration sizes evaluated:

n 10, 20, 50, 100, 200, 500

Unless otherwise specified, experiments use:

n = 100

Calibration samples are excluded from the final test set.

B.9 Reproducibility Details

To ensure reproducibility:

• All experiments use fixed random seeds. • Dataset splits are saved and reused across runs. • Hyperparameters are selected using validation data only. • Test environments remain completely unseen during training.

The complete preprocessing pipeline is identical across all baselines unless explicitly stated otherwise.

This ensures that observed performance differences arise from model design rather than preprocessing artifacts.

APPENDIX C
COMPLEXITY ANALYSIS

This appendix analyzes the computational and storage complexity of WiCaP during pre-training, calibration, and deployment.

C.1 Pre-Training Complexity

The Domain-Contrastive Encoder (DCE) is trained using supervised contrastive learning combined with domain-adversarial regularization.

Let:

Ns = number of source-domain training samples E = number of training epochs B = batch size d = embedding dimension

The dominant computational cost arises from pairwise similarity computation within each batch during supervised contrastive learning.

For a batch of size B, the similarity matrix requires:

$O(B^2)$

pairwise similarity evaluations.

Therefore, total training complexity is:

$O(Ns \cdot E \cdot B^2)$

Although the theoretical complexity scales quadratically with batch size, the computation is implemented efficiently through matrix multiplication operations on GPUs.

The environment classifier introduces negligible additional overhead because it consists of a small multilayer perceptron operating on d-dimensional embeddings.

In practice, the training cost is dominated by forward and backward passes through the CNN encoder backbone.

For the Widar 3.0 dataset, pre-training requires approximately 5–7 GPU hours on a single NVIDIA V100 or A100 GPU.

C.2 Density Ratio Estimation Complexity

Weighted conformal prediction requires estimation of the density ratio

w(x) = ptarget(x) / psource(x)

through a binary domain classifier.

Let:

Nc = number of unlabeled target calibration samples Ns$_c al = numberofsourcecalibrationsamples$

The binary classifier is trained on:

Ns$_c al + Nc$

samples.

Since the density-ratio estimator is a lightweight three-layer MLP, training complexity is:

O((Ns$_c al + Nc)EDR)$

where EDR denotes the number of density-ratio training epochs.

Because Nc is typically between 50 and 200 samples, this cost is negligible relative to encoder pre-training.

C.3 Conformal Calibration Complexity

Conformal calibration is performed once per deployment environment.

Let:

n = size of calibration set K = number of activity classes

The calibration procedure consists of:

Step 1: Compute nonconformity scores for all calibration samples.

Complexity:

$O(n \cdot K)$

Step 2: Sort calibration scores to obtain the conformal quantile.

Complexity:

$O(n \log n)$

Thus total calibration complexity is:

$O(nK + n \log n)$

Since n is small (typically 100), calibration completes within seconds on a standard CPU.

Unlike domain adaptation approaches, calibration requires no gradient updates to the main recognition model.

C.4 Inference Complexity

For a new CSI sample x, deployment proceeds as follows:

1. CSI preprocessing 2. Encoder forward pass 3. Class probability computation 4. RAPS score computation 5. Prediction set construction 6. Abstention decision

Let:

Fenc = complexity of encoder forward pass

The additional cost introduced by WiCaP beyond the base classifier is:

O(K)

because one nonconformity score is evaluated for each class.

Therefore total inference complexity is:

O(Fenc + K)

Since K is typically between 6 and 22 for current HAR benchmarks, the additional overhead is negligible.

C.5 Memory Requirements

WiCaP introduces almost no additional memory requirements beyond the base classifier.

Stored parameters include:

• Encoder parameters • Classification head parameters • Density-ratio estimator parameters (optional) • Conformal threshold qw • Abstention threshold set

The conformal layer itself introduces no trainable parameters.

Therefore memory complexity is:

O(——)

which is identical to the base recognition model.

C.6 Scalability Discussion

WiCaP scales favorably with increasing dataset size because:

1. Conformal calibration is performed only once per deployment environment.

2. Calibration complexity depends on calibration-set size n rather than total training size Ns.

3. Prediction-set construction scales linearly with the number of activity classes K.

4. No model fine-tuning is required after deployment.

Consequently, WiCaP preserves the computational efficiency of existing WiFi sensing systems while providing statistically valid uncertainty estimates and selective prediction capabilities.

Summary

Training Complexity: O(Ns · E · B²)

Density-Ratio Estimation: O((Ns$_{c}al + Nc)EDR$)

Calibration Complexity: O(nK + n log n)

Inference Complexity: O(Fenc + K)

Memory Complexity: O(——)

Theoretical overhead after deployment: Negligible relative to encoder inference.

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